

- 26** The ratio of turns of the primary coil to the secondary coil of a practical transformer is 10:1. The primary coil p.d. and current are 20 V and 3 A respectively. The efficiency of the transformer is 0.8.

Assuming that the energy loss is purely due to the resistance of the windings, which of the following statements is false?

A The secondary voltage is 2 V.

B The secondary current is 24 A.

C The output power is 60 W.

D The power loss is 12 W.

Ans: C

Option A: $N_p/N_s = V_p/V_s \rightarrow 10 = 20/V_s \rightarrow V_s = 2 \text{ V}$

Option B: $N_p/N_s = I_s/I_p \rightarrow 10 = I_s/3 \rightarrow I_s \text{ (ideal)} = 30 \text{ A.} \rightarrow I_s \text{ (efficiency} = 0.8) = 24 \text{ A}$

Option C: $P_s \text{ (output)} = V_s I_s = (2)(24) = 48 \text{ W}$

Option D: $P_s \text{ (loss)} = V_s I_s \text{ (loss)} = (2)(6) = 12 \text{ W}$